[Total No. of Question -04]

[Total No. of Pages - 01]

G H Raisoni International Skill Tech University, Pune

F.Y. B. Tech (TERM-II)

CAE I- summer 2026 (2025 Pattern)

DIGITAL LOGIC AND DESIGN (25UESL1105)

(Time: - 01 Hour) **(Max. Marks: 20)**

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	*Question*	Marks	CO	BL
Q1.(a)	Simplify and proof the following Boolean expressions using Boolean algebra: i. $\bar{A}BC + A\bar{B}C + AB\bar{C} + ABC = AB + AC + BC$ ii. $(A + B)(A + \bar{B}) = A$ write all the steps.	[5]	CO1	2
(b)	Perform the following operations on given decimal numbers using 2's complement method and verify your result: $(50)_{10} - (37)_{10}$	[5]	CO1	2
	OR			
Q 2.(a)	Given the binary number $(1011011)_2$ i. convert it into its equivalent Gray Code ii. find Excess-3 code of it. Write all the steps.	[5]	CO1	2
(b)	Represent the given decimal number $(396)_{10}$ in: i. Binary form ii. BCD code iii. Excess-3 code iv. Octal, and v. Hexa decimal form	[5]	CO1	2
Q 3. (a)	Determine the minimized expression in SOP form for the given logic function using K-map. Realize the expression in logical circuit form also showing all the steps: $Y(A, B, C, D) = \sum m(0,1,3,5,7,8,13,15)$	[5]	CO2	3
(b)	Design a 4×1 Multiplexer. Explain all the steps involved in designing the logic circuit. Also, derive the Boolean expression and draw the logic diagram.	[5]	CO2	2
	OR			
Q4. (a)	Given a Boolean function in Sum of Products (SOP) form $F(A, B, C)\sum m\ (0,1,2,5,8,9,10)$: i. Find the Product of Sums (POS) form of the given function. ii. Write the logical expression and draw the logic circuit corresponding to the SOP form.	[5]	CO2	3
(b)	Explain the operation of a Full Adder combinational circuit. Support your explanation with the truth table, Boolean expressions for Sum and Carry, and a logic circuit diagram.	[5]	CO2	2